

MC-AWS-DYNAMIC TABLES ASSIGNMENT PDF

--Q1

--Create a new database and schema to hold Dynamic tables.

```
CREATE OR REPLACE DATABASE DYNAMICDB;
```

```
CREATE OR REPLACE SCHEMA DTABLES;
```

--Q2

--Create a Dynamic table by joining the HRMS.HR.Departments and HRMS.HR.Locations table.

--Use the following properties and add all columns

--TARGET_LAG= 60 mins

--WAREHOUSE =COMPUTE_WH

--REFRESH_MODE=AUTO

--INITIALIZE=ON_SCHEDULE

```
CREATE OR REPLACE DYNAMIC TABLE DYT_DEPT_LOC
```

```
TARGET_LAG='60 minutes'
```

```
WAREHOUSE ='COMPUTE_WH'
```

```
REFRESH_MODE=AUTO
```

```
INITIALIZE=ON_SCHEDULE
```

```
AS
```

```
SELECT
```

```
    L.LOCATION_ID,  
    L.STREET_ADDRESS,  
    L.POSTAL_CODE,  
    L.CITY,  
    L.STATE_PROVINCE,  
    L.COUNTRY_ID,  
    D.DEPARTMENT_ID,  
    D.DEPARTMENT_NAME,  
    D.MANAGER_ID
```

```
FROM
```

```
    HRMS.HR.LOCATIONS L
```

```
    INNER JOIN HRMS.HR.DEPARTMENTS D ON L.LOCATION_ID = D.LOCATION_ID;
```

--Q3

--Select data from the Dynamic table and see if you can see the data

--If you see the data , how many rows are in the Dynamic table.

--If you are not able to see data what do you need to do to ensure that there is data in the DT and after that provide the row count

```
SELECT * FROM DYT_DEPT_LOC
```

```
ALTER DYNAMIC TABLE DYT_DEPT_LOC REFRESH;
```

```
SELECT * FROM DYT_DEPT_LOC
```

--Q4

--Check how many DT are in the schema , provide the count

--Review the properties of the DT and what is the value of SCHEDULING_STATE.

--Drop the DT and check how many DT are in the schema , provide the count

```
SHOW DYNAMIC TABLES;
```

```
DROP DYNAMIC TABLE DYT_DEPT_LOC;
```

```
SHOW DYNAMIC TABLES;
```

--Q5

--Re-create the DT by joining the HRMS.HR.Departments and HRMS.HR.Locations table.

--Use the following properties.

--TARGET_LAG= 60 mins

--WAREHOUSE =COMPUTE_WH

--REFRESH_MODE=AUTO

--INITIALIZE=ON_CREATE

--Select from the DT and provide the count, did you encounter any error, if no why, if yes why

CREATE OR REPLACE DYNAMIC TABLE DYT_DEPT_LOC

TARGET_LAG='60 minutes'

WAREHOUSE ='COMPUTE_WH'

REFRESH_MODE=AUTO

INITIALIZE=ON_CREATE

AS

SELECT

L.LOCATION_ID,
L.STREET_ADDRESS,
L.POSTAL_CODE,
L.CITY,
L.STATE_PROVINCE,
L.COUNTRY_ID,
D.DEPARTMENT_ID,
D.DEPARTMENT_NAME,
D.MANAGER_ID

FROM

HRMS.HR.LOCATIONS L

INNER JOIN HRMS.HR.DEPARTMENTS D ON L.LOCATION_ID = D.LOCATION_ID;

SELECT * FROM DYT_DEPT_LOC;

--Q6

--Update the HRMS.HR.Locations table for LOCATION_ID=1700 and update Street address from "2004 CHARADE RD" to "2024 CHARADE RD"

--After the update check the DT to see if the Street address has changed.if it has changed why , if not why ?

--If it has not changed how long will it take for it to show up in the Dynamic table.

--How can you propagate the change to the DT immediately.

--Once you propagate the changes , verify if the DT has been updated

```
UPDATE HRMS.HR.LOCATIONS
SET STREET_ADDRESS ='2024 CHARADE RD'
WHERE LOCATION_ID=1700;
```

```
SELECT * FROM DYT_DEPT_LOC;
```

```
ALTER DYNAMIC TABLE DYT_DEPT_LOC REFRESH;
```

```
SELECT * FROM DYT_DEPT_LOC;
```

--Q7

--Alter the Dynamic table to suspend it and check its scheduling state

```
SHOW DYNAMIC TABLES;
```

```
ALTER DYNAMIC TABLE DYT_DEPT_LOC SUSPEND;
```

```
SHOW DYNAMIC TABLES;
```

--Q8

--Alter the DT to update the Target Lag to 120 minutes

--Verify that it has changed

```
ALTER DYNAMIC TABLE DYT_DEPT_LOC SET TARGET_LAG = '120 minutes';
```

```
SHOW DYNAMIC TABLES;
```

--Q9

--Truncate the DT and report the outcome

```
TRUNCATE TABLE DYT_DEPT_LOC;
```

--Q10

--Create a table called NOT_A_REAL_TABLE with 1 column of type varchar

```
-- CREATE OR REPLACE TABLE NOT_A_REAL_TABLE
-- (
--   A varchar
-- );
--Create a DT with the below properties by selecting data on the table NOT_A_REAL_TABLE
--TARGET_LAG= Downstream
--WAREHOUSE =COMPUTE_WH
--REFRESH_MODE=AUTO
--INITIALIZE=ON_SCHEDULE
--After creation of the DT DROP the table NOT_A_REAL_TABLE.
--Next refresh the DT and check to see if there are any errors when refreshing it.
--Go to the INFORMATION_SCHEMA.DYNAMIC_TABLE_REFRESH_HISTORY and check to
see if the refresh has been recorded
```

```
CREATE OR REPLACE TABLE NOT_A_REAL_TABLE
(
  A varchar
);
```

```
CREATE OR REPLACE DYNAMIC TABLE DYT_NOT_A_REAL_TABLE
TARGET_LAG=DOWNSTREAM
WAREHOUSE ='COMPUTE_WH'
REFRESH_MODE=AUTO
INITIALIZE=ON_SCHEDULE
AS
SELECT * FROM NOT_A_REAL_TABLE
```

```
DROP TABLE NOT_A_REAL_TABLE;
```

```
ALTER DYNAMIC TABLE DYT_NOT_A_REAL_TABLE REFRESH;
```

```
SELECT CURRENT_TIMESTAMP
```

```
SELECT ---Update the TS below to yesterday's date
```

```
*
```

```
FROM
  TABLE (
```

```

        INFORMATION_SCHEMA.DYNAMIC_TABLE_REFRESH_HISTORY (
            NAME_PREFIX => 'DYNAMICDB.DTABLES',
            ERROR_ONLY => TRUE,
            DATA_TIMESTAMP_START => TO_TIMESTAMP_LTZ('2024-07-01 07:00:38',
'YYYY-MM-DD HH24:MI:SS')
        )
    );

```

--Create pipeline

--Q11

--Create a pipeline of DT, the TARGET_LAG for the last DT in the pipeline should be 60 mins.

--Choose other parameters such that the data is DT is populated immediately.

--1st DT = Join between Departments and Locations

--2nd DT = Join between 1st DT and Countries

--3rd DT = Join between 2nd DT and Regions

--Update data for location_id 1700 back from 2024 to 2003 and check if the data reflects in the 3rd DT.

--If not then refresh the DT and see the updated data

--1st DT = Join between Departments and Locations

CREATE OR REPLACE DYNAMIC TABLE DYT_DEPT_LOC

TARGET_LAG=DOWNSTREAM

WAREHOUSE ='COMPUTE_WH'

REFRESH_MODE=AUTO

INITIALIZE=ON_CREATE

AS

SELECT

```

    L.LOCATION_ID,
    L.STREET_ADDRESS,
    L.POSTAL_CODE,
    L.CITY,
    L.STATE_PROVINCE,
    L.COUNTRY_ID,
    D.DEPARTMENT_ID,
    D.DEPARTMENT_NAME,
    D.MANAGER_ID

```

FROM

```
HRMS.HR.LOCATIONS L
INNER JOIN HRMS.HR.DEPARTMENTS D ON L.LOCATION_ID = D.LOCATION_ID;
```

--2nd DT = Join between 1st DT and Countries

```
CREATE OR REPLACE DYNAMIC TABLE DYT_DEPT_LOC_COUNTRIES
TARGET_LAG=DOWNSTREAM
WAREHOUSE ='COMPUTE_WH'
REFRESH_MODE=AUTO
INITIALIZE=ON_CREATE
AS
SELECT
    DDL.LOCATION_ID,
    DDL.STREET_ADDRESS,
    DDL.POSTAL_CODE,
    DDL.CITY,
    DDL.STATE_PROVINCE,
    DDL.COUNTRY_ID,
    DDL.DEPARTMENT_ID,
    DDL.DEPARTMENT_NAME,
    DDL.MANAGER_ID,
    C.COUNTRY_NAME,
    C.REGION_ID
FROM DYT_DEPT_LOC DDL JOIN HRMS.HR.COUNTRIES C
ON DDL.COUNTRY_ID =C.COUNTRY_ID
```

```
ALTER DYNAMIC TABLE DYT_DEPT_LOC_COUNTRIES REFRESH;
```

--3rd DT = Join between 2nd DT and Regions

```
CREATE OR REPLACE DYNAMIC TABLE DYT_DEPT_LOC_COUNTRIES_REGION
TARGET_LAG=DOWNSTREAM
WAREHOUSE ='COMPUTE_WH'
REFRESH_MODE=AUTO
INITIALIZE=ON_CREATE
AS
SELECT
    DDLC.LOCATION_ID,
    DDLC.STREET_ADDRESS,
    DDLC.POSTAL_CODE,
    DDLC.CITY,
    DDLC.STATE_PROVINCE,
    DDLC.COUNTRY_ID,
    DDLC.DEPARTMENT_ID,
```

```
DDL.C.DEPARTMENT_NAME,  
DDL.C.MANAGER_ID,  
DDL.C.COUNTRY_NAME,  
R.REGION_ID,  
R.REGION_NAME  
FROM DYT_DEPT_LOC_COUNTRIES DDL.C JOIN HRMS.HR.REGIONS R  
ON DDL.C.REGION_ID =R.REGION_ID
```

```
SELECT * FROM DYT_DEPT_LOC_COUNTRIES_REGION;
```

```
UPDATE HRMS.HR.LOCATIONS  
SET STREET_ADDRESS ='2004 CHARADE RD'  
WHERE LOCATION_ID=1700;
```

```
SELECT * FROM DYT_DEPT_LOC_COUNTRIES_REGION;
```

```
ALTER DYNAMIC TABLE DYT_DEPT_LOC_COUNTRIES_REGION REFRESH;
```

```
SHOW DYNAMIC TABLES
```

```
ALTER DYNAMIC TABLE DYT_DEPT_LOC SUSPEND;  
ALTER DYNAMIC TABLE DYT_DEPT_LOC_COUNTRIES SUSPEND;  
ALTER DYNAMIC TABLE DYT_DEPT_LOC_COUNTRIES_REGION SUSPEND;  
ALTER DYNAMIC TABLE DYT_NOT_A_REAL_TABLE SUSPEND;
```

```
SHOW DYNAMIC TABLES
```